Supervised Learning Project: Income Classification

Sami Hnaien

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1 Introduction

In this project, we aim to apply supervised learning techniques to solve a binary classification problem: predicting whether an individual earns more than $50K per year.

This problem is widely studied in machine learning because it combines socio-economic variables with real-world prediction challenges.

The target variable is defined as:

- 0: Income $\leq 50K$
- 1: Income $> 50K$

Two classification models are implemented and compared in this study:

- Logistic Regression
- K-Nearest Neighbors (KNN)

The objective is not only to build predictive models but also to analyze their performance and understand their behavior on real data.

2 Dataset Description

The dataset used in this project is the Adult Income dataset, which contains demographic and economic information about individuals.

Each row represents an individual, described by several attributes such as age, education level, work status, and financial indicators.

For the purpose of this project, only numerical variables are selected in order to simplify the modeling process and ensure compatibility with distance-based algorithms such as KNN.

The selected features are:

- Age
- Education Number
- Hours per Week
- Capital Gain

- Capital Loss

The target variable is **income**, which is transformed into a binary variable:

- 0: income $\leq 50K$
- 1: income $> 50K$

This transformation allows the use of classification algorithms.

3 Objective of the Study

The main objective of this project is to compare the performance of two supervised learning models applied to income classification.

More specifically, the goals are:

- Build predictive models using Logistic Regression and KNN
- Evaluate model performance using multiple metrics
- Compare the strengths and weaknesses of each method
- Identify the most suitable model for this dataset

Additionally, this study aims to highlight the importance of proper data preprocessing and evaluation techniques in machine learning.

4 Methodology

4.1 Dataset Overview

Before applying machine learning models, it is important to understand the structure of the dataset.

The following figure presents a preview of the dataset, including the main variables and their format.

This preview highlights the presence of both numerical and categorical variables. In this project, only numerical features are selected for modeling.

```
## 'data.frame':    32561 obs. of  15 variables:
##  $ age           : int  39 50 38 53 28 37 49 52 31 42 ...
##  $ workclass     : chr  "State-gov" "Self-emp-not-inc" "Private
##  $ fnlwgt        : int  77516 83311 215646 234721 338409 28458
##  $ education     : chr  "Bachelors" "Bachelors" "HS-grad" "11th
##  $ education.num : int  13 13 9 7 13 14 5 9 14 13 ...
##  $ marital.status: chr  "Never-married" "Married-civ-spouse" "
##  $ occupation    : chr  "Adm-clerical" "Exec-managerial" "Hand
##  $ relationship  : chr  "Not-in-family" "Husband" "Not-in-fami
##  $ race          : chr  "White" "White" "White" "Black" ...
##  $ sex           : chr  "Male" "Male" "Male" "Male" ...
##  $ capital.gain  : int  2174 0 0 0 0 0 0 0 14084 5178 ...
##  $ capital.loss  : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ hours.per.week: int  40 13 40 40 40 40 16 45 50 40 ...
##  $ native.country: chr  "United-States" "United-States" "Unite
##  $ income        : chr  "<=50K" "<=50K" "<=50K" "<=50K" ...
```

Figure 1: Preview of the dataset structure

4.2 Exploratory Data Analysis (EDA)

Before applying machine learning models, an exploratory data analysis (EDA) was performed to better understand the dataset structure and the distribution of the target variable.

4.2.1 Target Variable Distribution

The target variable represents whether an individual earns more than 50K per year. It is transformed into a binary classification problem:

- 0: Income $\leq$ 50K
- 1: Income $>$ 50K

The distribution of the two classes is presented in Figure 2. The dataset shows an imbalance between the two categories, with a larger number of individuals belonging to the class with income less than or equal to 50K.

This imbalance may affect the performance of classification models, especially for metrics such as recall and F1-score. Therefore, different evaluation metrics are considered to provide a complete assessment of model performance.

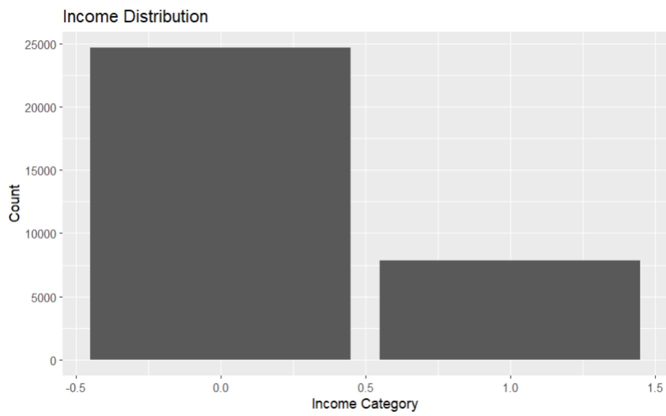

Figure 2: Distribution of the target variable (Income)

4.2.2 Correlation Analysis

A correlation analysis was performed on the numerical variables selected for the modeling process. The correlation matrix helps identify relationships between features and provides a better understanding of the dataset before applying machine learning algorithms.

The selected numerical variables are:

- Age
- Education Number
- Hours per Week
- Capital Gain
- Capital Loss

The correlation matrix is shown in Figure 3.

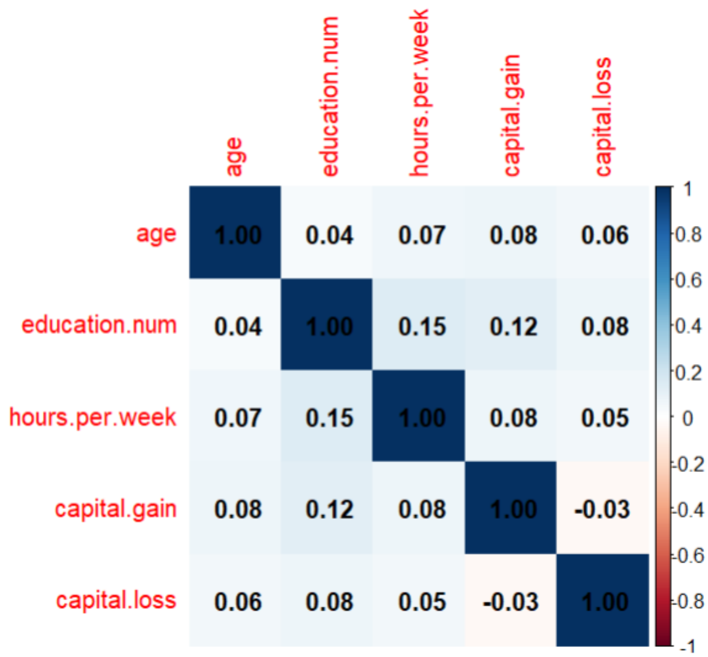

Figure 3: Correlation matrix of numerical features

4.3 Data Preprocessing

Data preprocessing is a crucial step in any machine learning project.

First, we select only numerical variables in order to simplify the analysis and ensure compatibility with the KNN algorithm, which is based on distance calculations.

The selected features are:

- Age
- Education Number
- Hours per Week
- Capital Gain
- Capital Loss

The target variable **income** is transformed into a binary variable:

- 0: income $\leq 50K$

- 1: income $> 50K$

4.4 Train-Test Split

To evaluate model performance, the dataset is divided into two subsets:

- Training set (70%): used to train the models
- Test set (30%): used to evaluate performance

This separation ensures that the models are evaluated on unseen data, which is essential to avoid overfitting.

4.5 Data Normalization

Normalization is particularly important for the KNN algorithm.

Since KNN relies on distance calculations, features with larger scales can dominate the results. To avoid this issue, Min-Max normalization is applied.

The normalization formula is:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

This transformation rescales all numerical variables to a range between 0 and 1.

4.6 Workflow Overview

The following figure summarizes the overall methodology used in this project. it includes data preprocessing, train-test splitting, normalization, model training, and evaluation.

4.7 Evaluation Metrics

To compare the performance of the models, several evaluation metrics are used:

- Accuracy: proportion of correctly classified observations
- Precision: proportion of true positives among predicted positives

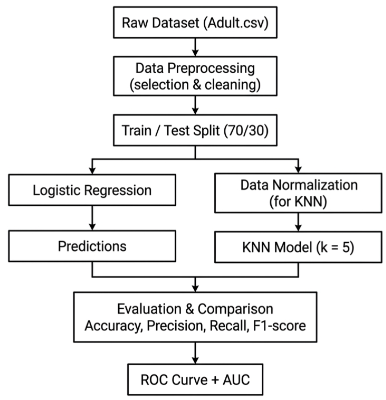

Figure 4: Overview of the data processing and modeling pipeline

- Recall: proportion of true positives among actual positives
- F1-score: harmonic mean of precision and recall
- ROC Curve and AUC: overall performance evaluation

These metrics provide a comprehensive assessment of model performance, especially in the presence of class imbalance.

5 Models and Results

5.1 Logistic Regression

Logistic Regression is a linear model used for binary classification. It estimates the probability that an observation belongs to a given class. The model is trained using the training dataset without normalization. After training, predictions are generated on the test set using a threshold of 0.5.

The model achieved the following accuracy:

- Accuracy: 0.81

5.2 K-Nearest Neighbors (KNN)

KNN is a non-parametric algorithm that classifies observations based on the majority class of their nearest neighbors. Unlike Logistic Regression, KNN requires normalized data to ensure that all features contribute equally to the distance calculation. Different values of k were tested, and the best performance was obtained around $k = 9$.

The model achieved the following accuracy:

- Accuracy: 0.83

5.3 Confusion Matrix and Metrics

To better understand model performance, confusion matrices are analyzed.

5.3.1 Logistic Regression

- Precision: 0.71
- Recall: 0.39
- F1-score: 0.51

```
[1] "Confusion matrix Log:"
         Actual
Predicted    0    1
        0 6957 1464
        1  390  958
```

Figure 5: Confusion Matrix for Logistic Regression

The confusion matrix shows that Logistic Regression struggles to correctly identify positive cases, which explains its relatively low recall.

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5.3.2 KNN

- Precision: 0.73
- Recall: 0.48
- F1-score: 0.58

```
[1] "Confusion matrix knn:"
          Actual
Predicted    0    1
        0 6911 1244
        1  436 1178
```

Figure 6: Confusion Matrix for KNN

KNN achieves better classification performance, particularly in detecting positive cases.

These results indicate that KNN outperforms Logistic Regression, especially in terms of recall and F1-score.

It is important to note that the dataset is imbalanced, with a higher proportion of individuals earning less than or equal to 50K.

This imbalance negatively impacts model performance, particularly for Logistic Regression, which exhibits a relatively low recall. This means that the model fails to correctly identify a significant number of high-income individuals (income ¿ 50K).

In contrast, KNN provides a better balance between precision and recall, which explains its higher F1-score.

5.4 K Selection Analysis

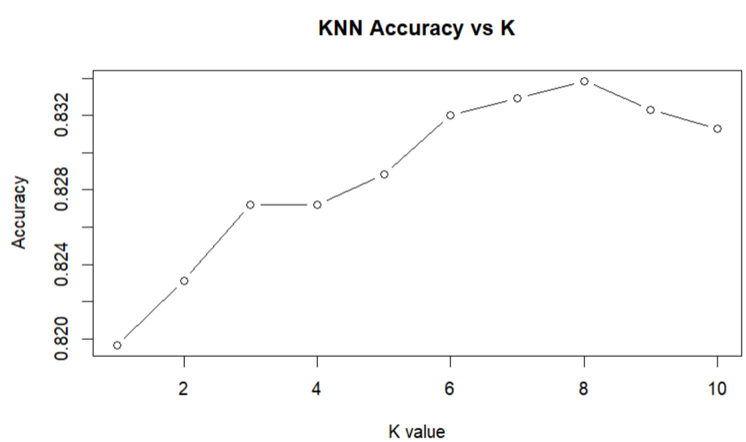

Figure 7: Accuracy of KNN for different values of k

The impact of the parameter k on KNN performance is analyzed. The figure shows that the model performance improves with increasing k up to a certain point, after which it stabilizes.
The choice of the parameter k plays a crucial role in the performance of the KNN model. A small k may lead to overfitting, while a large k may oversmooth the decision boundary.

5.5 ROC Curve and AUC

The ROC curve provides a visual representation of model performance.

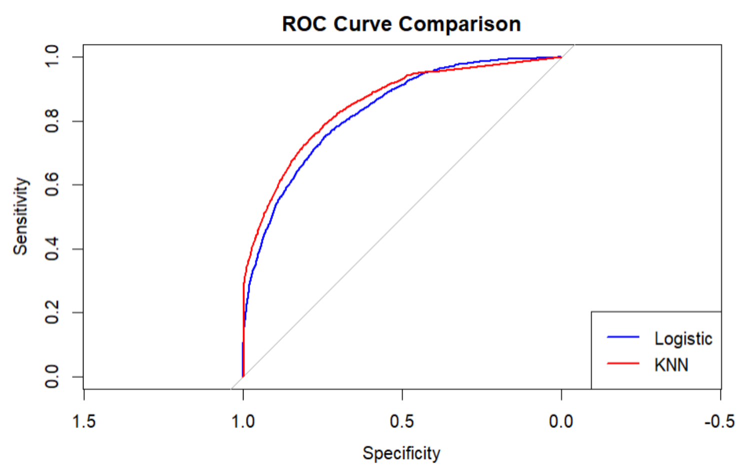

Figure 8: ROC Curve comparison between Logistic Regression and KNN

The AUC values are:

- Logistic Regression: 0.83
- KNN: 0.85

These results confirm that KNN slightly outperforms Logistic Regression in this classification task.

5.6 Discussion

Overall, both models perform well, but KNN shows better performance across most evaluation metrics.
This can be explained by the fact that KNN captures non-linear relationships in the data, while Logistic Regression is limited to linear decision boundaries. However, KNN is more computationally expensive and sensitive to the choice of k and data scaling.
On the other hand, Logistic Regression is simpler, faster, and easier to interpret.

6 Conclusion

In this project, we applied two supervised learning algorithms, Logistic Regression and K-Nearest Neighbors (KNN), to solve a binary classification problem using the Adult Income dataset.

The objective was to predict whether an individual earns more than $50K per year based on selected numerical features.

The results show that both models perform well, achieving good accuracy scores. However, KNN slightly outperforms Logistic Regression across several evaluation metrics, including accuracy, recall, F1-score, and AUC.

This difference can be explained by the ability of KNN to capture non-linear patterns in the data, whereas Logistic Regression assumes a linear relationship between the features and the target variable.

Despite its better performance, KNN has some limitations. It is computationally more expensive and highly sensitive to the choice of the parameter k and to feature scaling. In contrast, Logistic Regression is simpler, faster, and more interpretable.

6.1 Limitations

This study has some limitations:

- Only numerical features were used, while the dataset contains many informative categorical variables
- No advanced feature engineering was applied
- No cross-validation was performed to optimize hyperparameters

These limitations may affect the overall performance of the models.

6.2 Future Work

Several improvements can be considered for future work:

- Include categorical variables using encoding techniques
- Apply cross-validation to better tune model parameters
- Test more advanced models such as Decision Trees, Random Forest, or Neural Networks
- Perform feature selection to identify the most important variables

7 Appendix

The complete R code used in this project is provided separately.